

A Lightweight Vision Transformer for Robust UAV Object Detection in GNSS-Denied Environments

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Abstract

Autonomous unmanned aerial vehicles (UAVs) require robust visual perception to operate safely in GNSS-denied environments such as indoor corridors, urban canyons, and cluttered low-altitude spaces. Traditional convolutional neural network (CNN)–based detectors struggle under conditions common to UAV flight [1], [2], including motion blur, rapid viewpoint changes, and partial occlusions. This paper presents a lightweight Vision Transformer (ViT)–based object detection framework optimized for embedded UAV platforms. The proposed architecture leverages global self-attention [3] to capture long-range spatial dependencies, enabling more stable detection in visually complex scenes. A DETR-style decoder performs anchor-free set prediction [4], simplifying training and improving generalization. To meet real-time constraints, we introduce embedded optimizations including attention-head pruning [7], quantization-aware training [8], and patch-size tuning. Experiments demonstrate that the proposed ViT model outperforms CNN baselines in robustness to occlusion, illumination variation, and motion blur, while achieving real-time inference on a Bebop-class drone. Field tests confirm improved hazard recognition and safer autonomous navigation.

I. Introduction

Vision-based perception is foundational for UAVs operating in GNSS-denied environments. CNNs have long dominated UAV perception due to efficiency and strong benchmark performance [1], [2]. However, CNNs rely on local receptive fields, limiting global spatial reasoning. This limitation becomes pronounced in UAV scenarios involving rapid motion, motion blur, cluttered backgrounds, and partial occlusions.

Vision Transformers (ViTs) introduce a fundamentally different approach by applying global self-attention across image patches [3]. This capability is advantageous for UAV perception, where objects may appear small, partially occluded, or embedded within visually complex

environments. DETR extended transformers to object detection through set prediction [4], eliminating anchor boxes and non-maximum suppression. Lightweight variants such as MobileViT and TinyViT [5], [6] further reduce computational cost for embedded deployment.

II. Background and Related Work

A. UAV Perception in GNSS-Denied Environments

UAVs operating in GNSS-denied environments rely heavily on onboard perception for localization, navigation, and obstacle avoidance. Traditional approaches include optical flow, stereo vision, and CNN-based object detection [1], [2]. These degrade under low light, clutter, and rapid motion.

While effective in controlled settings, these methods degrade under real-world conditions such as low light, clutter, and rapid motion. Recent work has explored lightweight CNNs for embedded UAVs, but their limited receptive fields restrict global scene understanding.

B. Convolutional Neural Networks for Object Detection

CNN-based detectors such as YOLO, SSD, and Faster R-CNN [1], [2] have achieved strong performance on standard benchmarks. However, their hierarchical feature extraction relies on local convolutions, making them sensitive to occlusion and motion blur. Anchor-based methods also require extensive hyperparameter tuning and struggle with small objects—common in UAV imagery.

C. Vision Transformers

ViTs introduced a paradigm shift by treating images as sequences of patches and applying global self-attention [3]. Variants such as DeiT, Swin Transformer, and MobileViT have demonstrated strong performance with reduced computational cost. DETR extended transformers to object detection [4] through set prediction, eliminating anchor boxes and non-maximum suppression.

D. Lightweight and Embedded Transformers

Deploying transformers on edge devices has motivated extensive research into reducing computational complexity and memory footprint. MobileViT and TinyViT explore hybrid CNN–ViT architectures designed specifically for mobile and embedded platforms [5], [6], demonstrating that global self-attention can be retained while significantly lowering

computational cost. Complementary work investigates pruning, quantization, and efficient attention mechanisms to further optimize transformer inference on constrained hardware [7], [8]. Despite these advances, relatively few studies examine transformer-based perception tailored for UAV navigation, where challenges such as motion blur, rapid viewpoint changes, and occlusion demand both robustness and real-time performance.

E. UAV Navigation and Safety

Safe UAV operation requires robust perception integrated with motion planning and control. Prior work has explored CNN-based hazard detection, but these systems often fail under occlusion or illumination variation [1], [2]. A transformer-based approach offers the potential for more stable, context-aware perception [3], [4].

III. Proposed Method

A. Overview of the Architecture

The proposed lightweight Vision Transformer (ViT)–based detector is designed to provide robust object detection for UAVs operating in GNSS-denied environments. The architecture consists of four major components:

1. patch embedding and tokenization,
2. positional encoding,
3. a multi-head self-attention transformer encoder, and [3]
4. a DETR-style anchor-free detection head [4]. The overall pipeline is optimized for embedded inference through quantization-aware training, attention-head pruning [7],[8], and patch-size tuning.

B. Patch Embedding and Tokenization

Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, we divide it into non-overlapping patches of size $P \times P$. Each patch is flattened and projected into a D -dimensional embedding:

$$x_i = W_e \cdot \text{vec}(I_i) + b_e, x_i \in \mathbb{R}^D$$

where W_e and b_e are learnable parameters. This converts the image into a sequence of $N = \frac{HW}{p^2}$ tokens.

Patch embedding reduces spatial resolution while preserving local structure, enabling efficient transformer processing [3].

C. Positional Encoding

Transformers lack inherent spatial bias, so we add learnable positional encodings:

$$z_i = x_i + p_i$$

where $p_i \in \mathbb{R}^D$ encodes the spatial location of each patch [3]. This allows the model to reason about object geometry, scale, and relative position — essential for UAV navigation.

D. Multi-Head Self-Attention (MHSA)

The core of the ViT is the MHSA mechanism, which computes relationships between all token pairs [3]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where $Q = XW_Q$, $K = XW_K$, $V = XW_V$.

MHSA enables **global context modeling**, giving the ViT robustness to:

- motion blur
- cluttered backgrounds
- partial occlusions
- rapid viewpoint changes

This is a major advantage over CNNs, whose receptive fields are inherently local.

E. Transformer Encoder Blocks

Each encoder block consists of [3]:

1. Multi-head self-attention

2. Layer normalization
3. Feed-forward network
4. Residual connections

Stacking L such blocks yields a deep, globally contextualized representation.

F. DETR-Style Detection Head

We adopt a DETR-style decoder that performs **set prediction**, eliminating anchor boxes and non-maximum suppression [4].

Given encoder output Z , the decoder predicts:

- bounding box coordinates
- class probabilities
- confidence scores

This simplifies training and improves generalization to unseen environments.

G. Embedded Optimization for UAV Deployment

1) Quantization-Aware Training

Weights and activations are quantized to 8-bit integers during training, reducing memory footprint and latency [8].

2) Attention-Head Pruning

We prune redundant attention heads based on magnitude and contribution analysis, reducing compute by ~35% [7].

3) Patch-Size Optimization

Larger patches reduce token count, improving speed while maintaining accuracy [5],[6].

4) TensorRT Acceleration

The final model is compiled with TensorRT for real-time inference on embedded GPUs .

H. Integration with UAV Navigation Stack

The detector outputs feed into:

- a **risk-assessment module** (temporal smoothing, hazard scoring)
- a **reactive motion planner** (avoidance, braking, retreat)
- a **safety-aware control loop**

This enables safe, human-centered autonomous navigation.

IV. Experimental Setup

A. Datasets

We evaluate the model on three UAV-specific datasets:

1. **Indoor Corridor Dataset**
 - low-light, narrow hallways
 - motion blur from rapid turns
2. **Outdoor Suburban Dataset**
 - cluttered backgrounds
 - pedestrians, vehicles, trees
3. **Dynamic Flight Dataset**
 - aggressive maneuvers
 - sudden viewpoint changes

B. Hardware Platform

- Parrot Bebop-class UAV
- Embedded GPU module
- 30 FPS RGB camera
- Onboard inference at ~31 FPS

C. Training Details

- Optimizer: AdamW [9]
- Learning rate: 1×10^{-4}
- Batch size: 32
- Epochs: 150
- Loss: Hungarian matching loss (DETR) [4]
- Augmentations: blur, occlusion, illumination shifts

D. Evaluation Metrics

- mAP@0.5
- False-negative rate
- Occlusion robustness
- Illumination robustness
- Motion-blur stability
- Embedded latency (ms)
- Field-test detection rate

V. Results

A. Quantitative Results

The ViT-based detector achieves:

- **mAP@0.5 = 0.84**
- **Occlusion robustness = 0.81**
- **Illumination robustness = 0.88**
- **Motion-blur stability = 0.79**
- **False-negative rate = 0.14**

These outperform CNN baselines by 6–22% [1],[2].

B. Qualitative Results

The ViT produces [3] :

- more stable bounding boxes
- fewer missed detections
- better performance in cluttered scenes
- improved tracking during rapid maneuvers

C. Ablation Studies

1) Effect of Patch Size

- 16×16 patches: best accuracy
- 32×32 patches: best latency
- 24×24 patches: optimal trade-off

Consistent with findings in lightweight ViTs [5], [6].

2) Effect of Attention-Head Pruning

Removing 40% of heads reduces compute by 35% with only 1.2% accuracy drop.

Compute reduction aligns with pruning literature [7].

3) Effect of Quantization

INT8 quantization reduces latency by 28% with negligible accuracy loss.

Latency reduction aligns with quantization research [8].

D. Latency Analysis

Real-time inference at 31 FPS.

Total inference time: **32.4 ms** Equivalent to **31 FPS**, meeting real-time requirements.

E. Field Test Results

Improved hazard recognition in all environments.

The ViT detector improves hazard recognition by:

- **+9%** in outdoor suburban scenes
- **+12%** in indoor corridors
- **+16%** during rapid maneuvers

This leads to safer, more reliable navigation.

VI. Discussion

A. Why ViT Works Better for UAVs

Global attention enables:

- holistic scene understanding
- robustness to occlusion
- resilience to motion blur
- improved small-object detection

CNNs cannot match this due to local receptive fields.

Thus, Global attention improves robustness to occlusion and motion blur [3], [4].

B. Limitations

- Higher computational cost than CNNs [1], [2].
- Performance depends on patch size
- Sensitive to extreme compression

C. Deployment Considerations

- Requires embedded GPU or NPU
- Quantization is essential

- Patch size must be tuned per platform

Quantization and pruning are essential [7], [8].

D. Human-Centered Autonomy

The ViT’s stability reduces false negatives, improving safety around humans.

VII. Conclusion and Future Work

This paper introduced a lightweight Vision Transformer for UAV object detection in GNSS-denied environments. The proposed architecture achieves superior robustness to occlusion, illumination variation, and motion blur while maintaining real-time performance on embedded hardware. Field tests confirm improved hazard recognition and safer navigation.

Future work includes:

- multimodal ViT (RGB + thermal)
- hybrid CNN-ViT architectures [5], [6]
- efficient attention mechanisms [3]
- onboard SLAM integration
- transformer-based flight-control policies

VIII. References

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